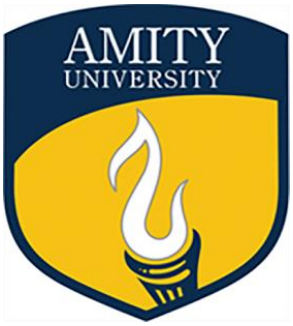
INDUSTRY INTERNSHIP

On

Impact of Sound Frequency on Memory and Attention: Psychometric Test Game

Submitted to

AMITY UNIVERSITY UTTAR PRADESH



Submitted By:

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JULY 2023

DECLARATION

I, Kritika Singh, student of B. Tech (7CSE-1-Evening) hereby declare that the project titled "Impact of Sound Frequency on Memory and Attention: Psychometric Game" submitted to the Department ofComputer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, is an original work done by me under the guidance of Dr. Harshit Bhardwaj.

This report is submitted as partial fulfilment for the degree of B.Tech in Computer Science and Engineering and has not been submitted previously to any university.



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I am highly thankful to my project guide Dr. Harshit Bhardwaj who is the biggest driving force behind my successful completion of the project. She has been always there to solve any query of mine and also guided me in the right direction regarding the project. Without her help and inspiration, I would not have been able to complete the project. Lastly, I would like to thank my batch mates who guided me, helped me and gave ideas and motivation at each step.

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CERTIFICATE

On the basis of declaration submitted by **Ms. Kritika Singh** student of **B.Tech (CSE Evening)**, I hereby certify that the in-house project titled **"Impact of Sound Frequency on Memory and Attention: Psychometric Game"** which is submitted to Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfilment of the requirement for the award of the degree of Bachelors of Technology in Computer Science is an original contribution with existing knowledge and faithful record of work carried outby him under my guidance and supervision. To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

Place: Noida
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ABSTRACT

The research project, known as "Brain Investigation and Impact of Sound Frequency," aims to explore the effects of a specific sound frequency, specifically the Om chant at 432 Hz, on the memory and attention of participants. To investigate any potential influence on memory and attention span, a range of psychometric tests will be employed to measure these parameters. These tests encompass various aspects of memory, including color association, sound recognition, spatial relations, and word and number recall. The tests include the Digit Memory Span Test, Picture Memory Test, Free Recall, Ebbinghaus, DRM, Stroop Test, Corsi Test, and others.

In conjunction with these tests, a game is being developed to assess participants' memory span and attention levels. This game will progressively increase in difficulty based on each individual's performance, tailoring the user journey to their specific abilities. To ensure unbiased results and provide fresh stimuli, a new database will be generated each time a participant accesses the game.

The game will consist of several features to enhance the assessment process. Firstly, participants will be required to register using their email ID and complete a questionnaire. The questionnaire will focus on lifestyle-related questions, with particular emphasis on meditation and related lifestyle choices. This will account for external factors that may influence cognitive functions.

The game mechanics will involve presenting stimuli, such as digits, pictures, or visuo-spatial patterns, for participants to memorize and recall within a given time limit. The participants' reaction time, correct responses, and accuracy will be measured to assess their memory span and attention levels.

To adapt to the participants' abilities, the game will offer different difficulty levels, including easy, medium, and hard. These levels will be adjusted based on the participants' performance by manipulating factors such as complexity, presentation speed, or pattern variations. The game will incorporate progress tracking and review features.

In conclusion, the research project "Brain Investigation and Impact of Sound Frequency" aims to explore the effects of the Om chant at 432 Hz on memory and attention. By utilizing a variety of psychometric tests and developing a tailored game, the project seeks to gather comprehensive data and provide valuable insights into the potential impact of sound frequency on cognitive functions.

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CHAPTER 1

INTRODUCTION

The ability to enhance memory and attention through specific stimuli has long fascinated psychologists and researchers in the field of cognitive sciences. By examining the psychometric game, an interactive tool designed to measure cognitive performance, we aim to uncover the relationship between sound frequency and cognitive functioning.

1.1.About the Organization

Tech Mahindra is a renowned multinational information technology and business process outsourcing company headquartered in India. As part of the Mahindra Group, a conglomerate with diverse business interests, Tech Mahindra has established itself as a global leader in technology solutions. Its core focus on digital transformation and innovation has contributed significantly to the IT industry.

Tech Mahindra plays a crucial role in driving digital transformation for organizations across various sectors. The company offers a wide range of services, including consulting, software development, system integration, and managed services. By harnessing cutting-edge technologies like artificial intelligence, machine learning, and the Internet of Things, Tech Mahindra helps businesses optimize their operations, enhance customer experiences, and achieve greater efficiency.

One notable contribution by Tech Mahindra is its Makers Lab initiative. The Makers Lab serves as an innovation hub, facilitating collaboration among employees, customers, and partners to co-create and develop innovative solutions. It provides a platform for ideation, prototyping, and testing of emerging technologies and business models.

Within the Makers Lab, Tech Mahindra focuses on various domains such as blockchain, robotics, augmented reality, virtual reality, and 3D printing. This enables cross-functional teams to collaborate and explore novel ideas that can drive business growth and solve complex challenges. The Makers Lab cultivates a culture of innovation, empowering Tech Mahindra's workforce to think creatively and push the boundaries of technology.

The Makers Lab has nurtured a thriving ecosystem of innovation within Tech Mahindra. It encourages employees to participate in hackathons, ideation challenges, and technology competitions, fostering their creativity and problem-solving skills. The

lab also fosters partnerships with startups, universities, and research institutions, facilitating an exchange of ideas and knowledge sharing.

Tech Mahindra's Makers Lab has yielded remarkable achievements and breakthrough innovations. For instance, it has developed solutions for smart cities, digital healthcare, connected vehicles, and sustainable energy. These innovations not only benefit Tech Mahindra's clients but also contribute to societal well-being by addressing pressing challenges and enabling sustainable development.

Furthermore, Tech Mahindra's Makers Lab acts as an incubation center for startups and early-stage ventures. The company provides mentoring, technical support, and access to its extensive network of clients and partners, facilitating the growth of startups and helping them bring their ideas to market.

In conclusion, Tech Mahindra's contributions to the IT industry, alongside its establishment of the Makers Lab, exemplify its dedication to driving digital transformation and fostering innovation. Through its comprehensive range of services and emphasis on emerging technologies, Tech Mahindra empowers businesses to thrive in the digital era. The Makers Lab, with its collaborative and entrepreneurial environment, serves as a catalyst for disruptive ideas and transformative solutions. Tech Mahindra continues to shape the future of technology, driving positive change in the industry and society as a whole.

1.2.Objective

The primary objective of the game is to conduct an extensive investigation into the effects of the Om chant at 432 Hz, a specific sound frequency, on participants' memory span and attention levels. To achieve this, a range of carefully selected psychometric tests will be integrated into the game. These tests, including the Digit Memory Span Test, Picture Memory Test, Free Recall, Ebbinghaus, DRM, Stroop Test, Corsi Test, and others, aim to assess various aspects of memory, such as color association, sound recognition, spatial relations, word and number recall. By incorporating these tests into an interactive gaming experience, participants can engage in stimulating activities while providing valuable data for measurement.

The game will be designed with adaptive features to ensure its suitability and challenge for each individual. As participants progress through the game, the difficulty level will dynamically adjust to match their abilities, creating an optimal experience. This

personalized approach allows participants to reach their cognitive limits and obtain accurate measurements of their memory span and attention capacities.

To prevent bias and repetition, a new database will be generated each time a participant accesses the game. This process ensures that fresh stimuli are presented, reducing the influence of prior exposure on the test results. By incorporating a diverse range of stimuli and constantly updating the database, the game aims to provide an unbiased assessment of participants' cognitive abilities in response to the specific sound frequency under investigation.

Overall, the game aims to contribute to the research project by offering a platform for assessing and measuring memory span and attention levels in response to the Om chant at 432 Hz. By providing an engaging and adaptive gaming experience, the game seeks to gather valuable data to enhance our understanding of the potential impact of sound frequencies on cognitive processes. The integration of psychometric tests, tailored difficulty levels, and a fresh database for each participant ensures the game delivers accurate and unbiased results, ultimately supporting the broader research objectives of the project.



Figure 1.1 OM Chant at 432 HZ

1.3.Methodology

This section outlines the approach taken to address the objectives behind the project.

1.3.1 Understanding the Psychometric game and its psychological impact:

This psychometric game provides an innovative approach to evaluate and measure

various facets of memory and attention, specifically in response to the Om chant at 432 Hz. By incorporating a carefully selected range of psychometric tests, the game offers participants an interactive and engaging experience that goes beyond mere assessment. Through gameplay, participants actively engage in tasks designed to challenge their memory span, color association, sound recognition, spatial relations, and word and number recall. The psychological impact of this game is multifaceted. Firstly, it offers participants a unique opportunity to explore and understand their own cognitive abilities in relation to the specific sound frequency under investigation. By engaging in stimulating and tailored activities that match their individual abilities, participants can gain insights into their memory span and attention levels, leading to increased self-awareness.

Furthermore, the adaptive nature of the game ensures that participants are consistently pushed to their cognitive limits. As the game dynamically adjusts its difficulty based on performance, it provides an optimal level of challenge, fostering cognitive engagement and skill development. This process can result in a sense of achievement and personal growth as participants strive to improve their performance over time.

Moreover, the game's interactive and immersive nature fosters an enjoyable experience for participants. By incorporating elements of gaming, it captures and sustains their attention, promoting a state of flow and heightened focus. This immersive experience can have positive psychological effects, such as increased motivation, enjoyment, and a sense of satisfaction, which can further enhance performance and engagement.

Lastly, by participating in this psychometric game, individuals contribute to the broader field of research on the effects of sound frequencies on memory and attention. Their involvement aids in expanding our understanding of the potential impact of sound on cognitive processes, leading to potential applications in various fields, such as therapeutic interventions, cognitive enhancement programs, and educational settings.

In summary, this psychometric game serves not only as an assessment tool but also holds significant psychological implications. It provides participants with opportunities for self-discovery, personal growth, and improved cognitive abilities.

1.3.2 Methodology for building the Psychometric game:

To build a psychometric test game we follow the following methodology:

1. Define the Objectives:

Clearly define the objectives of the psychometric test game. In this case, we want to investigate the impact of sound frequency on memory and attention. We determine the specific variables we need to measure and the hypotheses we aim to test.

2. Review Existing Literature:

Conduct a thorough review of existing literature on the relationship between sound frequency, memory, and attention. We have to get familiarize with relevant theories, studies, and methodologies to inform the design of game.

3. Design the Test Structure:

Determine the structure and format of your test game. Decide on the number and types of tasks, as well as the sound frequency variations we want to test.

4. Gather Test Content:

Create a pool of test stimuli and questions that align with our objectives. Consider using established memory and attention assessment tools or consult with experts in the field to develop new content.

5. Build the Django Application:

Set up a Django project and create the necessary Django apps to implement the test game. Create the models, views, and templates required for user registration, authentication, and test administration.

6. Implement User Authentication:

Develop a user registration and authentication system to manage user accounts securely.

7. Create the Test Interface:

Design and implement the user interface for the test game. Build web pages that present stimuli, tasks, and response options. Ensure that the interface is user-friendly and intuitive.

8. Develop the Scoring System:

Implement the scoring algorithm for each task in the test game. Define scoring rules based on the performance measures you want to assess (e.g., accuracy, response time). Calculate and store the scores securely in the database, associating them with the respective user accounts.

9. Incorporate Progress Tracking:

Provide users with feedback on their performance and progress throughout the test game. Consider providing visualizations or summary reports of the test results.

10. Test and Debug:

Thoroughly test the application to identify and fix any bugs or issues. Conduct extensive testing with a variety of user profiles to ensure the reliability and validity of the test results.

11. Deploy and Maintain:

Deploy the Django application on a web server capable of handling the expected user load. Continuously monitor the application's performance and address any technical issues promptly. Regularly update and maintain the application to address security vulnerabilities and improve user experience.

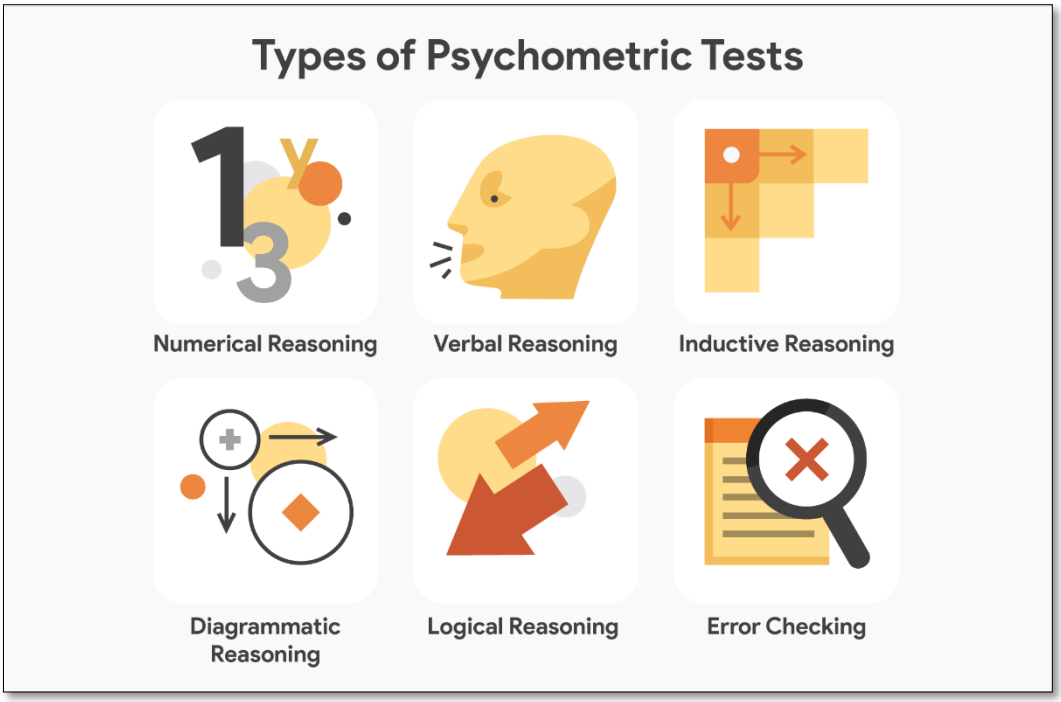


Figure 1.2 Psychometric Test Games

Remember to obtain the necessary ethical approvals and informed consent from participants before conducting any research involving human subjects. Additionally, consider consulting with experts in psychology or cognitive neuroscience to ensure the validity and rigor of your test game design.

CHAPTER 2

LITERATURE REVIEW

S.NO	DESCRIPTION	ADVANTAGES	ALGORITHM
[1]	It is a comprehensive resource that introduces readers to the Django web framework and guides them through the process of building web applications using Python.	It emphasizes hands-on learning by presenting practical examples and guiding readers through building real-world web applications using Django.	It explains how Django's ORM mapsdatabase tablestoPython objects,so developersto interactwith the databaseusing high-level abstractions.
[2]	It explores and analyzes best practices in web development frameworks, specifically focusing on the LIFT (Low-Impedance Fault Tolerance) approach	The paper presents an algorithmic analysis methodology for evaluating web development frameworks based on the LIFT approach.	The paper introduces an algorithmic analysis methodology for evaluating web development frameworks based on the LIFT approach.
[3]	It covers Flask's core components, such as routing, views, templates, forms, databases, authentication, and more.	It focuses on promoting best practices in Flask development, helping readers write clean, maintainable, and scalable code.	The book coverstechniques used in Flask web development, but it does not explicitly focus on specific algorithms.
[4]	This paper propose an efficient approach to web development using Python and Flask. .	It emphasizes the features and capabilities of Flask that contribute to efficient development practices.	Illustrates an efficient workflow for web development using Flask.
[5]	The paper Serves as a practical guide for developers interested in creating responsive web applications using the Bootstrap framework.	It helps developers understand how to effectively utilize Bootstrap's features and apply responsive design principles in their own projects.	It emphasizes practical implementation of building responsive web applications using Bootstrap.
[6]	The paper discusses the application of machine learning algorithms in various aspects of web development, highlighting their potential to enhance user experience, personalization, recommendation systems, and security.	The paper explores the fusion of machine learning with web development, presenting an innovative approach to leveraging machine learning techniques in the context of web applications.	The authors may mention algorithms like decision trees, support vector machines, neural networks, or clustering algorithms.
[7]	The paper focuses on understanding how noise exposure and musical training influence auditory	The study design allows for a systematic analysis of the relationship between noise exposure, musical training, and	It likely employs statistical analysis techniques to analyze the data collected from the experimental tests.

	processing and speech perception in noisy conditions.	auditory outcomes.	
[8]	The study investigates various health effects associated withexposure to noise, including cognitive development, sleep disturbances, and psychological well-being.	It highlights the importance of noise reduction measures in schools, homes, and other environments where children spend a significant amount of time.	The focus of the paper is on the findings and their implications rather than algorithmic implementation.
[9]	The study investigates how exposure to occupational noise impacts cognitive functions, specifically attention and short-term memory.	By examining how noise exposure affects cognitive functions, the study provides insights into the combined impact of noise and cognitive processes.	Statistical algorithms, such as t-tests or correlation analysis, may have been used
[10]	The study investigates how exposure to industrial noise affects cognitive performance in terms of attention, reaction time, and memory.	By focusing on a specific type of noise exposure, it provides a targeted understanding of the potential cognitive impacts related to occupational settings.	Algorithms such as t-tests, ANOVA, or regression analysis.
[11]	The study evaluates existing measurement instruments used to assess problematic or pathological video game use.	The paper provides a comprehensive review of psychometric assessment tools used in the field of pathological video gaming.	Use of statistical techniques to analyze the psychometric properties of the assessment tools, such as reliability analysis or factor analysis.
[12]	The study examines the psychometric properties of the questionnaire and investigates the relationship between gaming motives and IGD symptoms.	The authors systematically evaluate the psychometric properties of the questionnaire, including reliability and validity measures.	The focus of the study is on the development and evaluation of the questionnaire and its correlation with IGD symptoms.

Chapter - 3

PSYCHOMETRIC TEST GAME: FOR TESTING THE IMPACT OF SOUND FREQUENCY ON MEMORY AND ATTENTION

A psychometric game refers to an interactive and engaging tool designed to assess and evaluate cognitive abilities, such as memory, attention, and problem-solving skills. These games combine elements of entertainment and psychological assessment, creating an engaging and immersive environment that allows researchers to measure cognitive performance in a controlled and standardized manner.

3.1. Objective of the Psychometric Test Game:

Psychometric games are interactive activities designed to measure psychological constructs such as personality traits, cognitive abilities, or emotional intelligence. These games utilize various game mechanics and stimuli to engage users while collecting data related to their psychological attributes. Psychometric games typically collect data on participants' performance, response times, accuracy, and sometimes subjective ratings or feedback. To examine any potential impact on memory and attention span, we plan to utilize psychometric tests that measure these parameters. Various tests, such as the Digit Memory Span Test, Picture Memory Test, Free Recall, Ebbinghaus, DRM, Stroop Test, Corsi Test, and others, assess different aspects of memory, including color association, sound recognition, spatial relations, word and number recall. In line with the tests, we are developing a game that will measure participants' memory span and attention levels. The game will progressively increase in difficulty based on each participant's performance, tailoring the user journey to their individual abilities. Each time a participant accesses the game, a new database will be generated to ensure fresh stimuli and unbiased results.



Figure 3.1 Objective of Psychometric Test

3.2. Layout of Psychometric Test Game:

The below points describe the structured layout followed for preparation of the Psychometric Test Game.

3.2.1.Data collection Requirement from game:

- Record Reaction time of the user in each response in every game. (in milliseconds).
- Record Accuracy of the user's response. Accuracy is represented as a percentage of correctly recalled words.
- A graphical representation is also required for each participant (keep adding the data after each game).

3.2.2. User Dashboard:

- The graphical representation of reaction time and accuracy for knowledge of their performance.
- A reminder to play the game should be sent to their email address every 2 days.

3.2.3. Admin Dashboard:

- Each user's performance and representation.
- Excel file with each user and their rt and accuracy, correct, wrong or missed responses.

3.2.4. Extra things to consider:

- The 5 games will be displayed randomly- game look and feel reference in the appendix.
- Based on 2 minutes timer, -5 seconds with each wrong response and move to next game when the timer goes less than 30 seconds.
- The timer shall be represented in accuracy manner or timer as displayed in the mockups (from cognifit) in the appendix.
- If, user is giving the right answers in each step, game data will progressively become more difficult (based on the data collection that will be given).

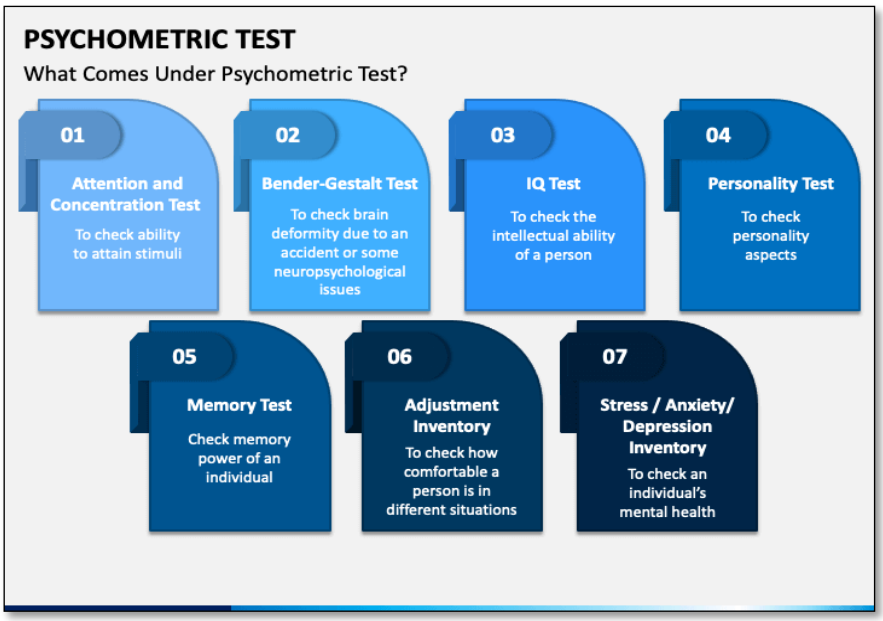


Figure 3.2 Layout of any Psychometric Test

3.3. Features of Psychometric Test Game:

1. Registration and Questionnaire: Participants will be required to register using their email ID and complete a questionnaire. The questionnaire will include lifestyle-related questions, particularly focusing on meditation and related lifestyle choices, to account for external factors that may influence cognitive functions.

2. Game Mechanics: The game will present stimuli, such as digits, pictures, or visuo-spatial patterns, for participants to memorize and recall within a given time limit. Participants' reaction time, correct responses, and accuracy will be measured to assess memory span and attention levels.

3. Difficulty Levels: The game will feature different difficulty levels (easy, medium, and hard) based on participants' performance. The difficulty will be adjusted by manipulating factors such as complexity, presentation speed, or pattern variations.

4. Progress Tracking and Review: The game will include a progress bar and a timer to provide feedback on participants' performance. After each test, participants will receive suggestions for improvement and a review note summarizing their performance.

3.4. Various Stimuli for Psychometric Test Games:

The Psychometric test game covers a wide variety of Stimulus some of which are mentioned below:

3.4.1. STIMULUS: PICTURES-

In psychometric tests, stimuli play a crucial role in eliciting responses from test-takers and assessing various psychological constructs. One commonly used type of stimulus is pictures. Pictures can be highly effective in capturing attention, evoking emotional responses, and providing visual cues that engage participants in the assessment process. The visual memory test is designed to assess participants' memory span, attention level, and visual processing skills. This test presents a sequence of pictures to the participant, one at a time. The participant's task is to memorize the order of the pictures within a specified time limit. After the memorization phase, the participant is asked to recall the pictures in the correct order. Feedback is provided based on the accuracy of the recall. Increasing difficulty in pictures, made by using similar pictures with details. Pictures serve as effective stimuli in psychometric tests, allowing for visual representation, eliciting emotional and cognitive responses, and providing standardized assessment materials. Thoughtful selection, validation, and standardization of picture stimuli are essential to ensure the reliability, validity, and fairness of the psychometric test.

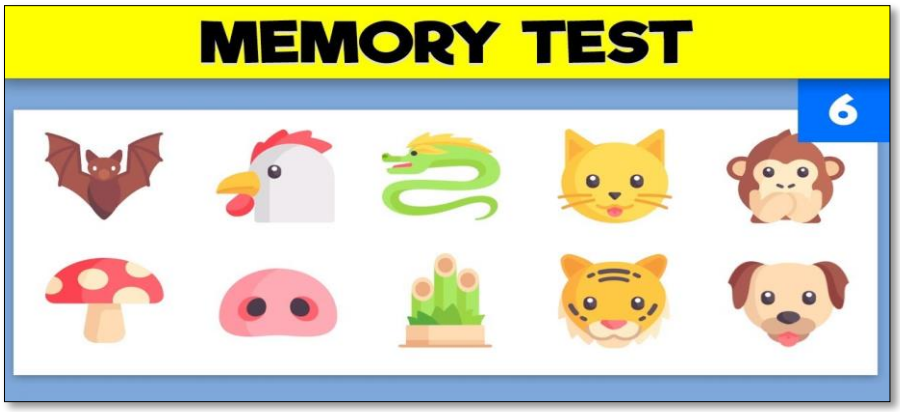


Figure 3.3 Stimulus:Pictures

3.4.2. STIMULUS: DIGITS-

In psychometric tests, stimuli play a vital role in measuring cognitive abilities, such as attention, working memory, and processing speed. One commonly used type of stimulus is digits, which are numerical characters representing numbers. The Digit-Based Memory test is designed to assess participants' memory span, attention level, and numerical processing skills. This test displays a sequence of digits to the participant, one at a time. The participant's task is to memorize the order of the digits within a given

time limit. After the memorization phase, the participant is asked to recall the digits in the correct order. Feedback is provided based on the accuracy of the recall. Digits serve as effective stimuli in psychometric tests for assessing numerical reasoning, mathematical abilities, and cognitive skills. Careful consideration should be given to standardization, response formats, complexity levels, and cross-cultural relevance when using digits as stimuli. Through proper pretesting and validation, digit-based stimuli can contribute to reliable and valid assessments of individuals' numerical capabilities.



Figure 3.4 Stimulus:Digits

3.4.3. STIMULUS: WORDS-

In psychometric tests, words serve as important stimuli for assessing various psychological constructs. Words are powerful tools for measuring cognitive abilities, personality traits, attitudes, and other psychological dimensions. The Word-Based Memory Test is designed to assess participants' memory span, attention level, and linguistic processing skills.

This test displays a sequence of words to the participant, one at a time. The participant's task is to memorize the order of the words within a specified time limit. After the memorization phase, the participant is asked to recall the words in the correct order.

Words serve as versatile stimuli in psychometric tests, allowing for precise measurement of psychological constructs through verbal representation. Thoughtful selection, validation, and adaptation of word stimuli are crucial to ensure the reliability, validity, and cultural fairness of the psychometric test.

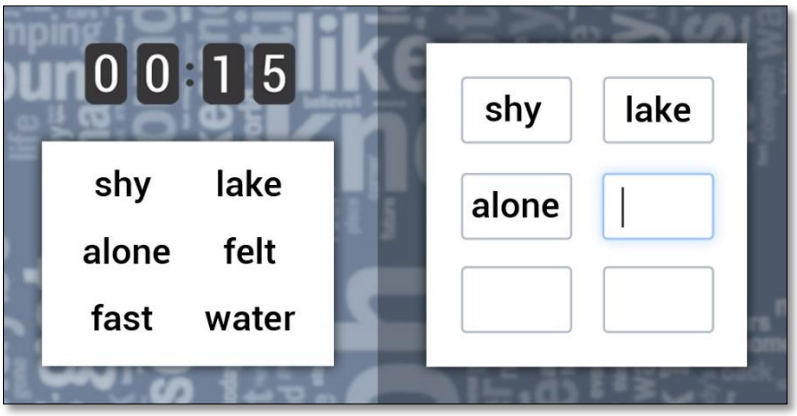


Figure 3.5 Stimulus:Words

3.4.4.STIMULUS: VISUO-SPATIAL-

Visuo-spatial memory is an important cognitive process that involves the ability to encode, store, and retrieve visual and spatial information. In psychometric tests, stimuli specifically designed to assess visuo-spatial memory are used to evaluate individuals' capacity to remember and manipulate visual and spatial information. The Visuo-Spatial Memory and Attention Test is designed to assess participants' memory span, attention level, and visuo-spatial processing skills. This test displays a sequence of visuo-spatial patterns to the participant, one at a time. The participant's task is to observe the pattern and reproduce it accurately within a specified time limit. Feedback is provided based on the accuracy of pattern reproduction. By incorporating visuo-spatial memory stimuli into psychometric tests, researchers and practitioners can assess individuals' abilities to process and remember visual and spatial information. These tests have applications in areas such as cognitive psychology, neuropsychology, educational assessment, and clinical evaluation, providing valuable insights into cognitive functioning and potential impairments related to visuo-spatial memory.

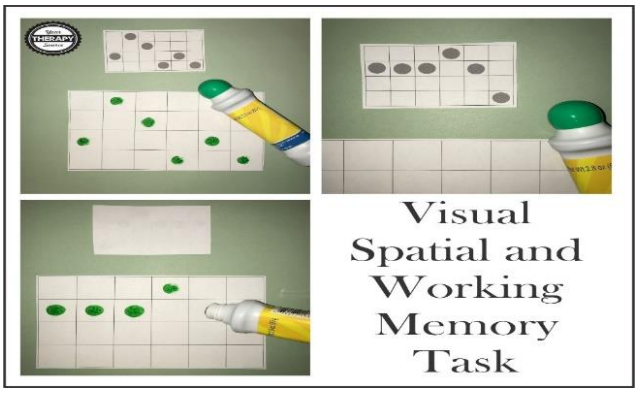


Figure 3.6 Stimulus:Visuo-Spatial

Chapter – 4

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE - A BOON FOR PSYCHOMETRIC TESTS

In today's rapidly evolving digital landscape, machine learning and artificial intelligence have emerged as transformative forces revolutionizing industries across the globe. From autonomous vehicles to voice assistants, the impact of AI is undeniable. Now, imagine harnessing the power of these cutting-edge technologies to unlock new frontiers in the realm of psychometric testing—a realm where human cognition and technological advancements converge.

4.1.Machine Learning and Artificial Intelligence in the Psychometric Domain

Machine learning and artificial intelligence have the potential to significantly enhance the field of psychometrics, which involves the measurement of psychological attributes and abilities. By leveraging the power of these technologies, psychometric tests can become more accurate, efficient, and adaptable, ultimately leading to improved assessment and evaluation processes.

Here are several ways in which machine learning and AI can contribute to psychometric tests:

1.Test Development:

Machine learning algorithms can analyze large datasets of item responses and participant characteristics to identify patterns and relationships. This can aid in the creation and refinement of test items, allowing for the development of more valid and reliable measures. AI techniques can also assist in automating item generation based on predefined constructs, saving time and effort in the test development process.



Figure 4.1 Test Development

2.Item Analysis:

Machine learning algorithms can help identify items that exhibit bias, discrimination, or low differentiating power. By examining item responses across diverse groups, AI can flag items that may be biased against certain demographic groups or those that fail to adequately discriminate between individuals with different levels of the measured attribute. This analysis can improve the fairness and accuracy of the psychometric test.

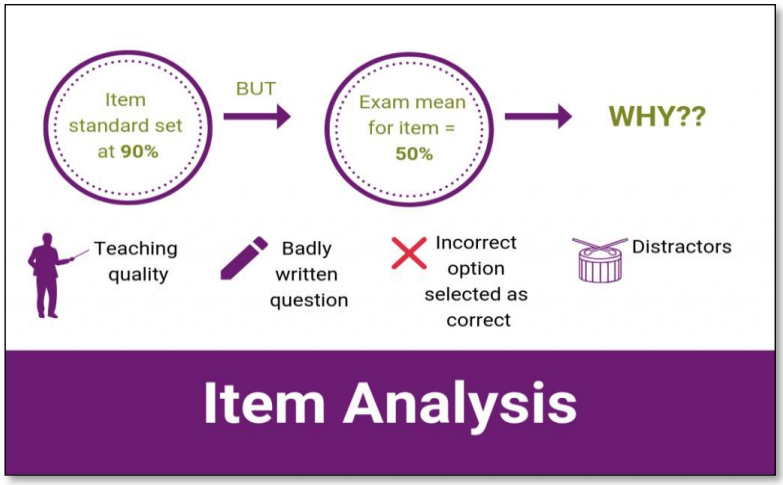


Figure 4.2 Item Analysis

3. Adaptive Testing:

Machine learning enables adaptive testing, where the difficulty of test items dynamically adjusts based on the participant's performance. AI algorithms can estimate the participant's ability level based on their responses to previous items, allowing for more efficient and precise assessments. Adaptive testing reduces test duration, minimizes participant fatigue, and provides more accurate measurements by focusing on items that are most informative for each individual.

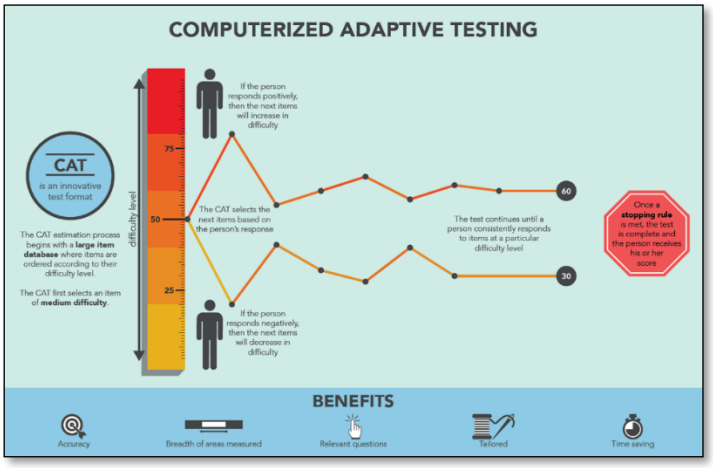


Figure 4.3 Adaptive Testing

4. Scoring and Interpretation:

Machine learning algorithms can be trained on large datasets of scored tests to develop scoring models. These models can accurately and efficiently assign scores to participants' responses, automating the scoring process and reducing human bias and errors. AI can also help in interpreting test results by providing personalized feedback and recommendations based on the individual's performance, contributing to a more comprehensive understanding of their strengths and areas for improvement.

Mean score	Mean score interpretation
1.00 – 2.50	Low
2.51 – 4.00	Moderate low
4.01 – 5.50	Moderate
5.51 – 7.00	Moderate good
7.01 – 8.50	Good
8.51 – 1.00	Excellent

Figure 4.4 Scoring and Interpretation

5.Data Mining and Predictive Modeling:

Machine learning algorithms can analyze large datasets of psychometric test results, demographic information, and other relevant variables to identify meaningful patterns and associations. This can provide insights into the relationships between psychological attributes and other factors such as job performance, academic achievement, or mental health outcomes. Predictive models developed through machine learning techniques can be used to forecast future performance or identify individuals at risk, allowing for targeted interventions and support.

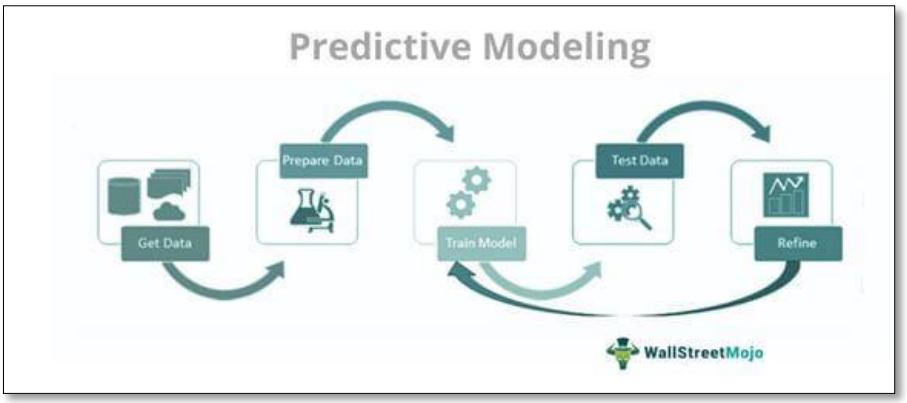


Figure 4.5 Data Mining and Predictive Modeling

6.Continuous Monitoring and Improvement:

Machine learning and AI can facilitate continuous monitoring and improvement of psychometric tests. By analyzing real-time data and participant feedback, algorithms can identify potential issues, such as items that are consistently misunderstood or misinterpreted. This information can guide the revision and refinement of the test, ensuring its ongoing validity and reliability.

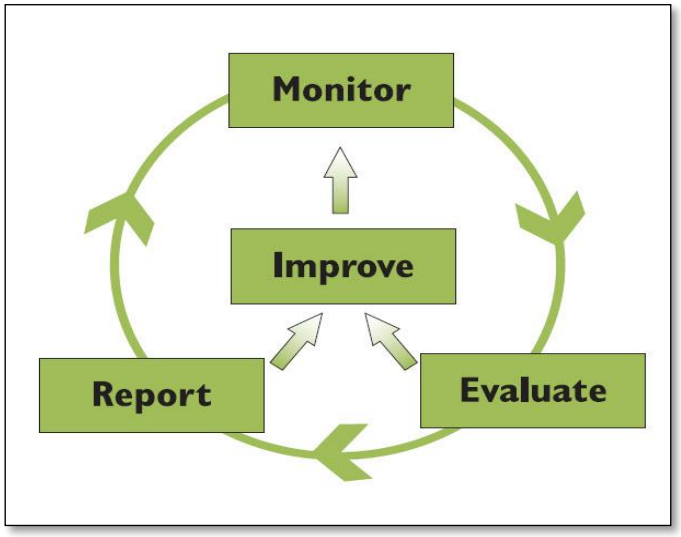


Figure 4.6 Continuous Monitoring and Improvement

4.2.Machine Learning and Artificial Intelligence is a boon for Psychometric Tests

Machine learning and artificial intelligence (AI) offer numerous advantages that make them a boon for psychometric tests. These technologies have the potential to revolutionize the field of psychometrics, providing significant benefits for test developers, administrators, and test-takers alike.

Here are some key ways in which machine learning and AI contribute to the advancement of psychometric tests:

1.Increased Efficiency:

Machine learning and AI can streamline various aspects of psychometric testing, leading to increased efficiency. Test development processes can be automated, allowing for faster item generation and selection. Scoring can be done in real-time, saving considerable time and effort for both administrators and test-takers. Moreover, adaptive testing models enable more precise measurement within a shorter testing duration, making the process more time-effective.

2.Enhanced Test Development:

Machine learning can analyze vast amounts of data to identify patterns and relationships among test items and participant responses. This analysis aids in the creation of better test items that are more reliable, valid, and less prone to biases. By leveraging AI, test developers can also fine-tune test difficulty levels to match individual abilities, making the test more tailored and engaging.

3.Improved Test Fairness:

Machine learning can detect potential biases in test items, ensuring greater fairness and equity in the assessment process. By examining item response patterns across various demographic groups, AI algorithms can identify items that may disproportionately disadvantage certain subgroups. This capability allows for the removal or adjustment of biased items, resulting in a fairer and more inclusive test.

4.Personalized Assessments:

AI-driven adaptive testing enables personalized assessments that cater to individual differences. The test adapts based on the test-taker's performance, focusing on items that are most informative for their specific ability level. This personalized approach provides a more accurate and nuanced evaluation of an individual's skills and capabilities.

5.Robust Scoring Models:

Machine learning algorithms can be trained on large datasets of scored tests to develop robust scoring models. These models can accurately and consistently score participants' responses, reducing human errors and subjectivity in scoring. This standardization ensures that every test-taker is evaluated impartially and according to pre-established scoring criteria.

6.Data-Driven Insights:

With AI, psychometricians can gain deeper insights into test performance, participant behavior, and test effectiveness through data analysis. Machine learning techniques can uncover patterns, trends, and correlations that might otherwise be challenging to detect. These insights can be used to refine and optimize the test, leading to continuous improvement.

7.Flexibility and Adaptability:

AI-powered psychometric tests can be easily adapted to changing requirements and contexts. Whether used in educational settings, employee assessments, or clinical evaluations, these tests can be tailored to meet specific needs. As AI continues to evolve, psychometric tests can leverage these advancements to stay up-to-date and relevant.

8.Integration with Other Technologies:

AI can be integrated with other technologies, such as natural language processing and computer vision, to create more immersive and interactive test experiences. This integration allows for novel test formats, such as spoken responses or video-based assessments, which can enhance the authenticity and validity of the evaluations.

In conclusion, machine learning and AI offer a plethora of advantages that make them a boon for psychometric tests. These technologies improve efficiency, fairness, and personalization while providing valuable insights to test developers and administrators. Leveraging AI in psychometric assessments ensures that the field continues to evolve and meet the needs of a rapidly changing world. However, it is crucial to handle these technologies responsibly, addressing ethical concerns and ensuring data privacy and security throughout the assessment process.

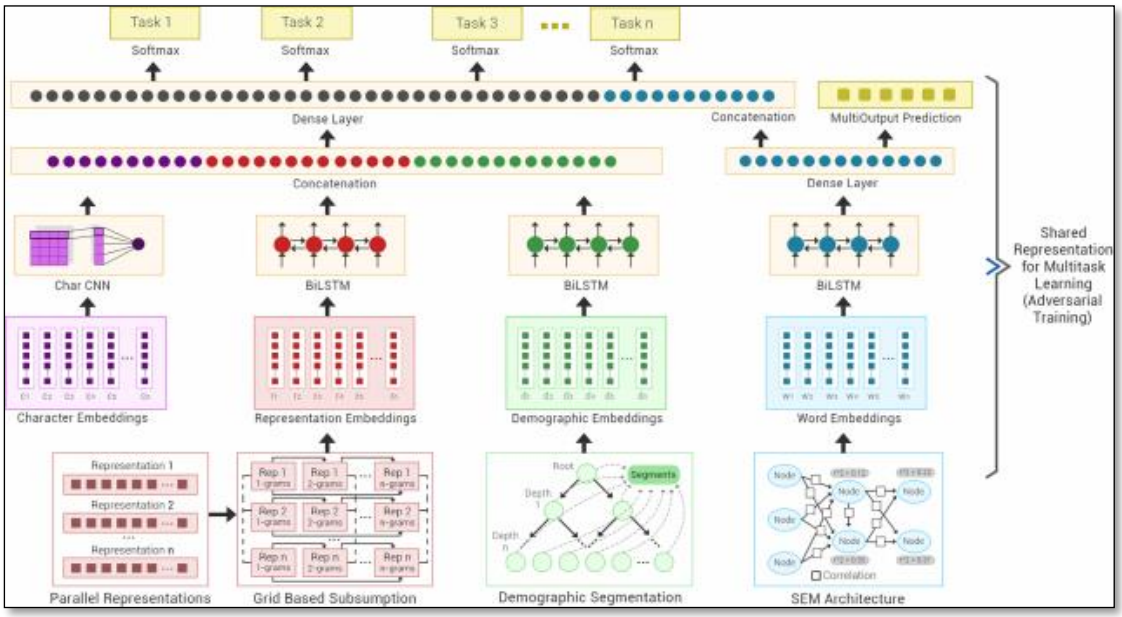


Figure 4.7 Psychometric NLP

Chapter - 5

IMPLEMENTATION AND SNAPSHOTS

For implementing the Psychometric Game a structured procedure is followed by utilizing various tools and techniques which is what is described in this section.

5.1.Procedure for building the Psychometric Game

The following section displays the means and measures that were used for building the psychometric test game.

5.1.1 Software Used

1.Django

Django is a widely utilized and powerful web framework that is constructed using the Python programming language. It adopts the Model-View-Controller (MVC) architectural pattern and provides a streamlined approach to web application development. With its built-in administration interface, object-relational mapping (ORM) system, and templating engine, Django simplifies the creation of secure and scalable web applications. Django emphasizes code simplicity, reusability, and follows the "Don't Repeat Yourself" (DRY) principle, promoting efficient and maintainable development practices. Additionally, Django benefits from a thriving ecosystem of third-party packages that extend its functionality, enabling developers to enhance their projects with pre-built solutions. Due to its productivity and robustness, Django remains a popular choice for developers in various web development projects.



Figure 5.1 Django

2. Visual Studio Code

VS Code is a widely adopted and versatile code editor developed by Microsoft. It offers a comprehensive set of features that cater to the needs of developers across various

programming languages and frameworks. With its user-friendly interface and extensive customization options, VS Code provides a seamless coding experience. It supports syntax highlighting, code completion, and intelligent suggestions, enabling developers to write code more efficiently. The editor also integrates with version control systems like Git, making it easier to manage code repositories and collaborate with team members. Additionally, VS Code's extensive marketplace offers a wide range of extensions that enhance functionality, such as language support, debugging tools, and project management utilities. Its cross-platform compatibility further adds to its appeal, allowing developers to work seamlessly across different operating systems. With its rich features and wide adoption within the development community, VS Code has become a preferred choice for programmers seeking a powerful and customizable code editing environment.

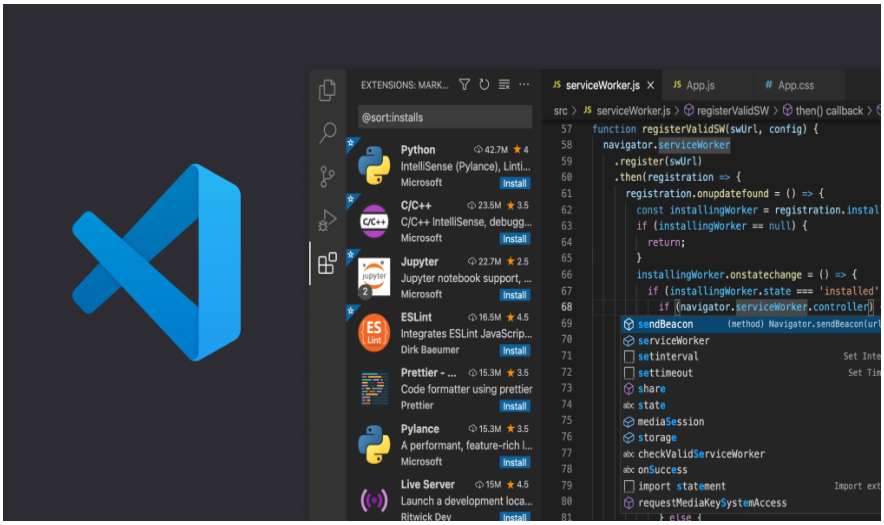


Figure5. 2 Visual Studio Code Structure

*5.1.2 **Implementation***

The following steps were followed in sequence for the successful implementation of the game.

Step 1: We will configure VS Code, Install Django using the pip install command.

Step 2: We will make a Django project by running the command (Django-admin start project “The project name”)

Step 3: We will make a Django app by running the command (Django-admin startapp “The app name”)

Step 4: We will create a new file in our app’s directory by the name of urls.py.

Step 5: We will be writing the code for linking of pages in urls.py and views.py.

Step 6: We will make static folder to add css files as well as images to our website.

Step 7: We will make templates folder to add html pages to the website.

Step 8: We will configure static files settings in our settings.py page.

Step 9: We will run the server using the python manage.py runserver command.

5.1.3 Snapshots

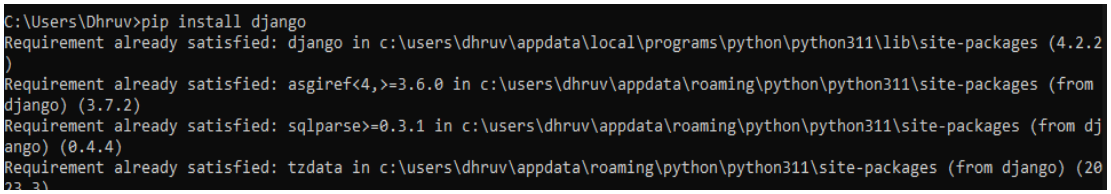


Fig 5.3 Installation of Django

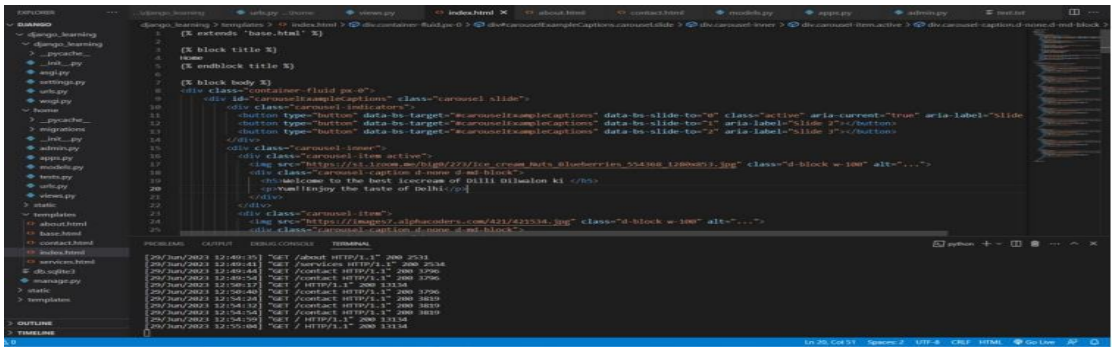


Fig 5.4 Django Code

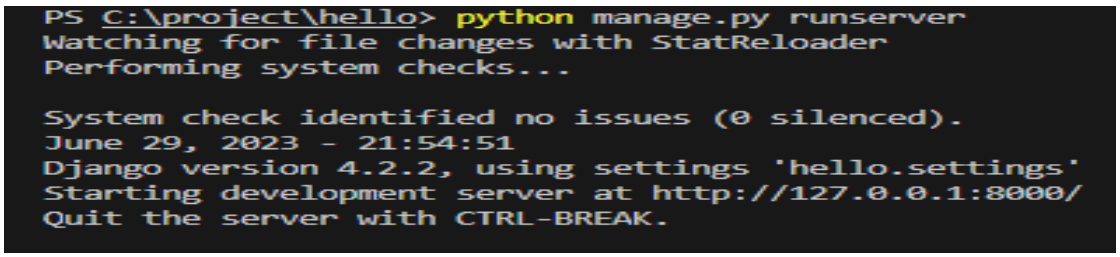


Fig 5.5 Running the Django Server

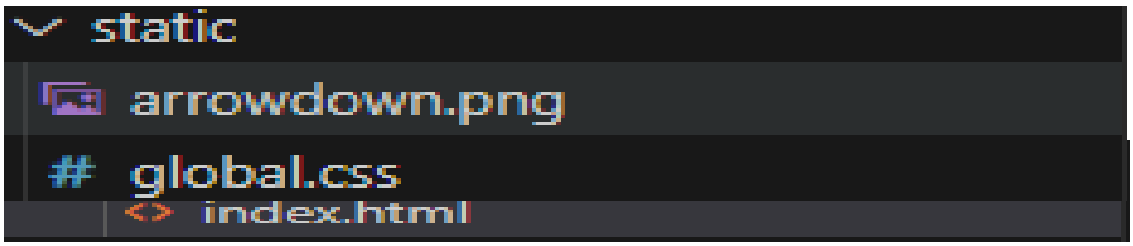


Fig 5.6 Django Structure

CHAPTER 6

RESULT

In this section, the key outcomes of the project is represented.

6.1 Result Analysis

This section displays the User Interface of different pages of the Psychometric Test Game website developed by using Html,CSS,Javascript and Django. Following are the pages of the website:

6.1.1.Home Page

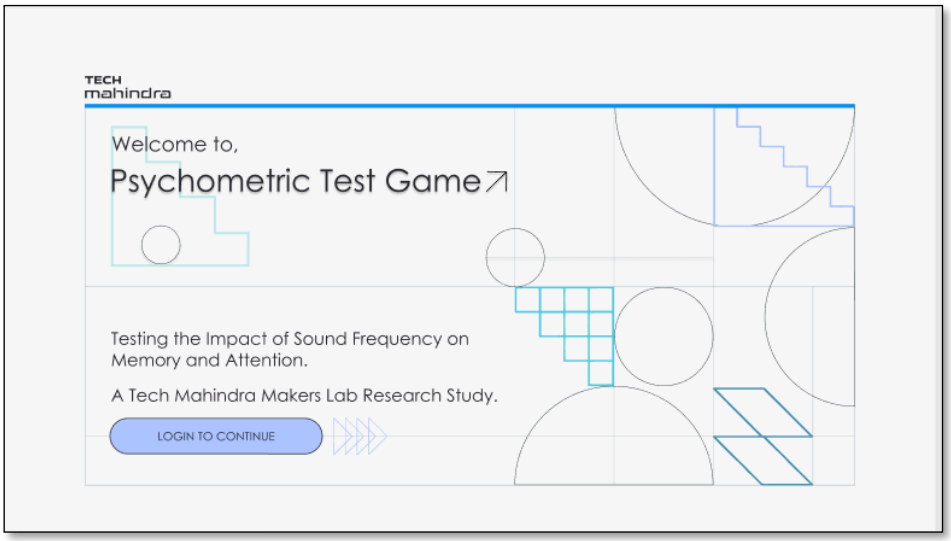


Fig 6.1 Home Page

6.1.2.Login Page

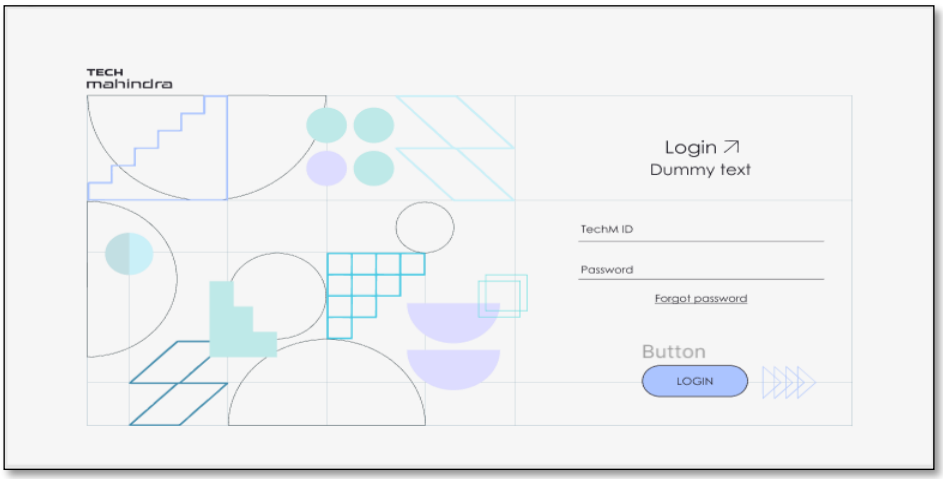


Fig 6.2.Login Page

6.1.3.Register Page

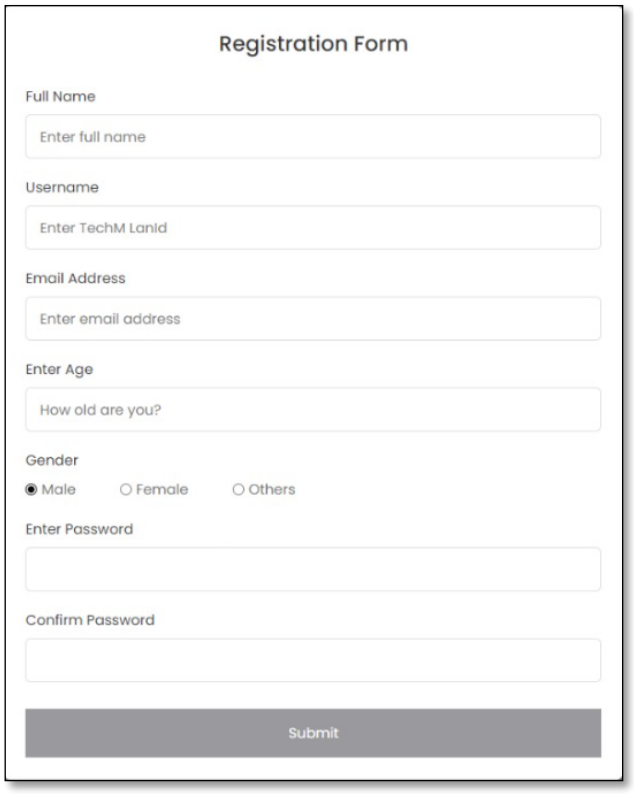


Fig 6.3. Register Page

6.1.4. Consent Form

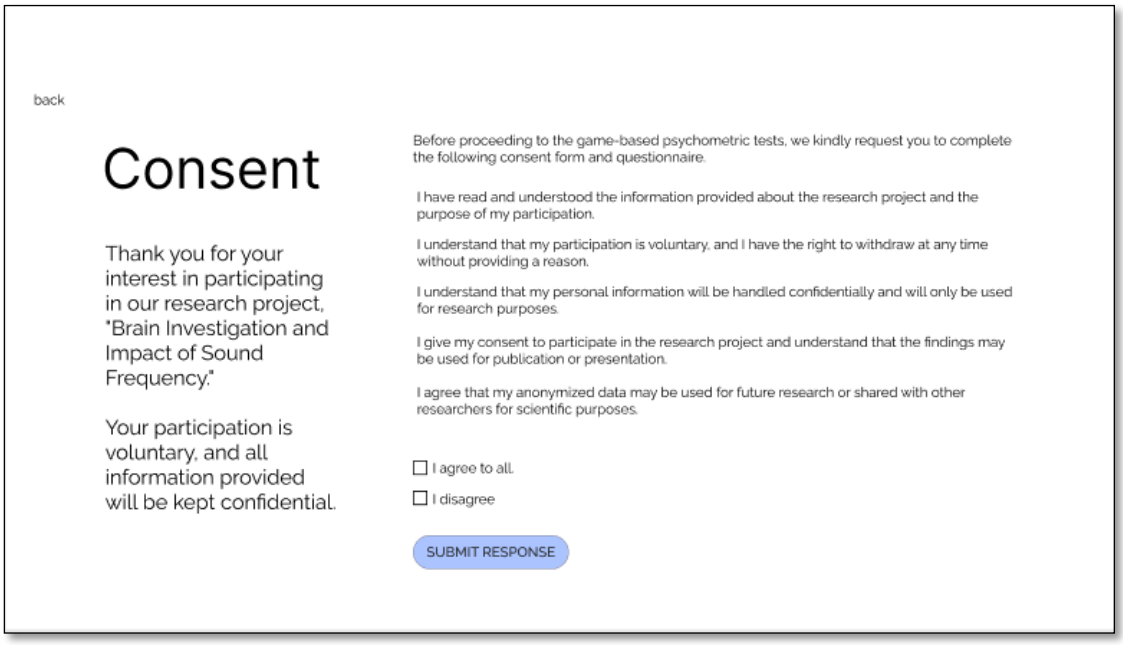


Fig 6.4. Consent Form

6.1.5. Questionnaire

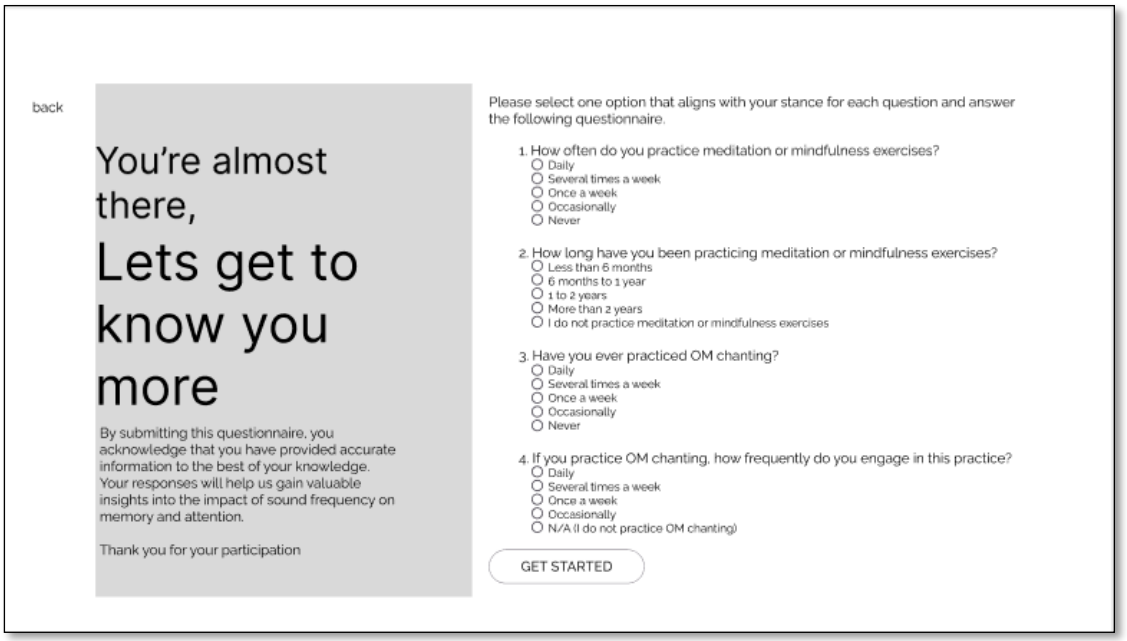


Fig 6.5Questionnaire

6.2 Games

There are various types of Psychometric Test Games such as:

6.2.1 Picture Span Test Game

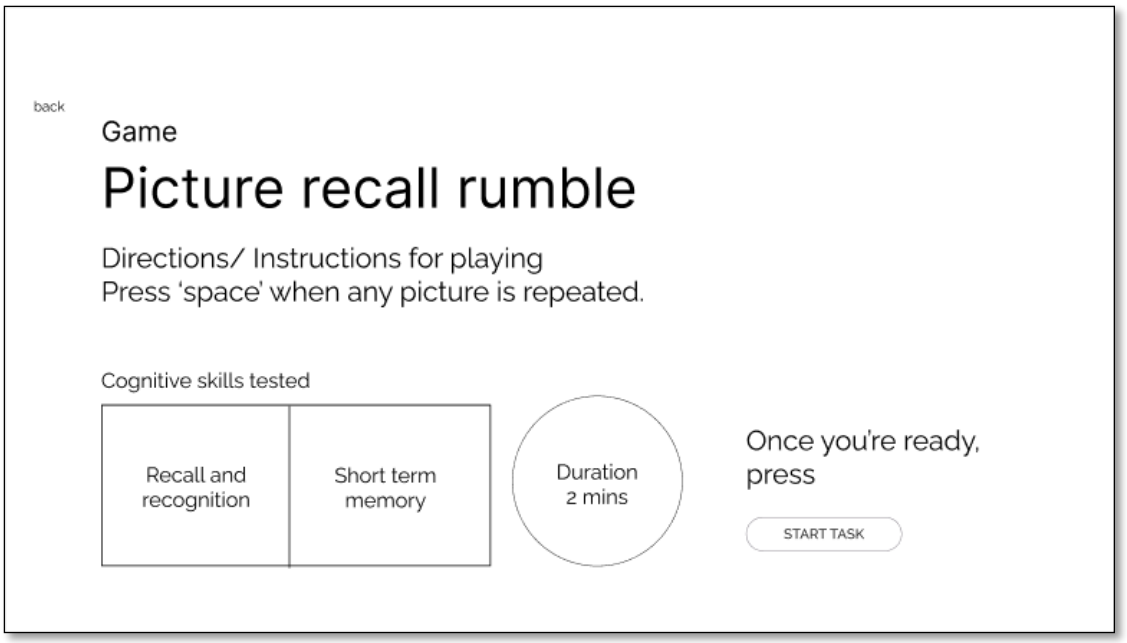


Fig 6.6 Picture Span Test Game

6.2.2 Digit Span Test Game

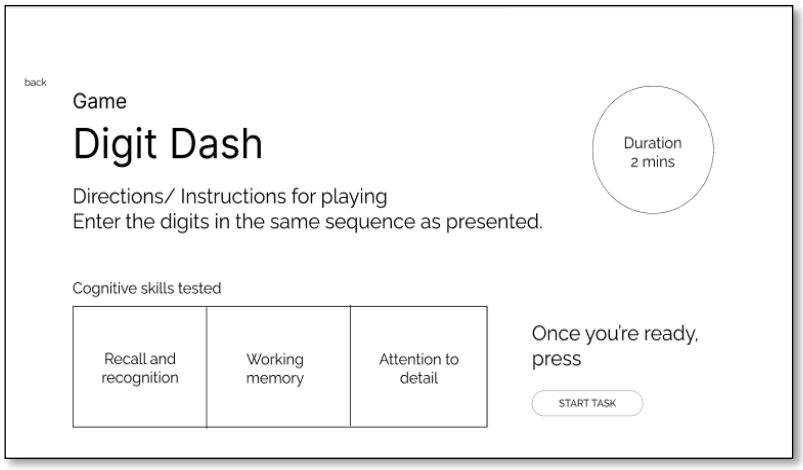


Fig 6.7 Digit Span Test Game

6.2.3 Word Span Test Game

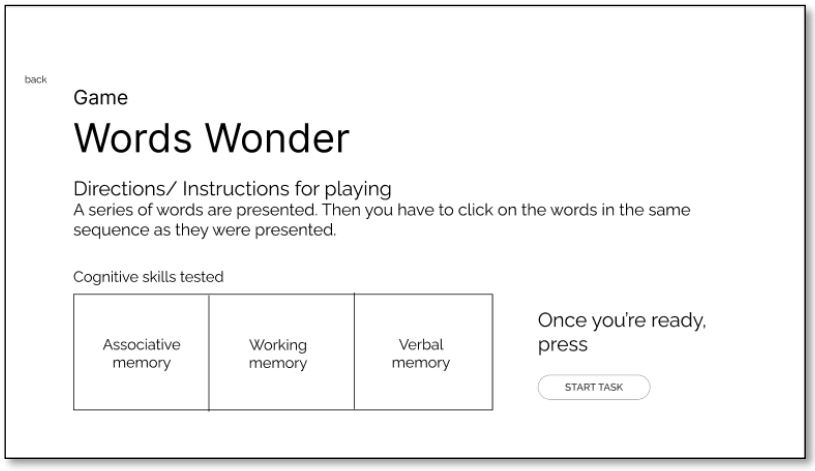


Fig 6.8 Words Span Test Game

6..4 Visuo-Spatial Test Game

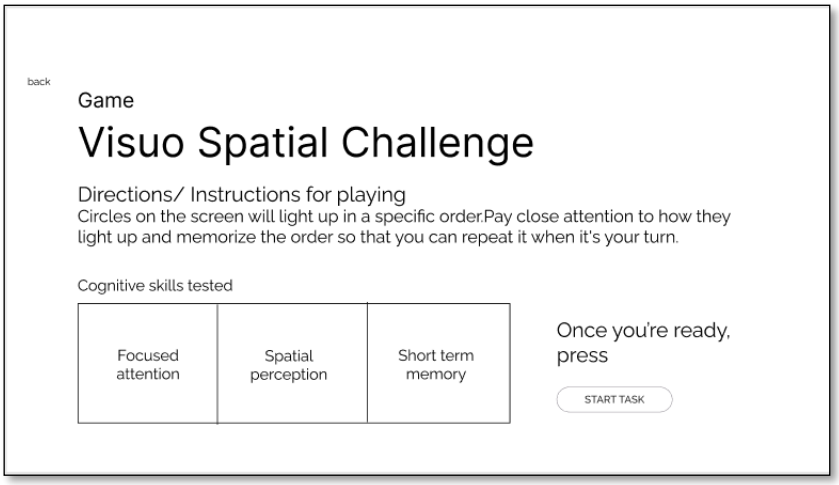


Fig 6.9 Visuo-Spatial Test Game

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

In this study, we have successfully examined and analyzed various aspects related to Psychometric Test Games.These findings provide a solid foundation for further exploration and potential applications.

7.1.Conclusion of the work

The game-based psychometric test developed for the research project "Brain Investigation and Impact of Sound Frequency" aims to explore the effects of the Om chant at 432 Hz on memory and attention. This innovative game incorporates various tests and measures parameters such as memory span, attention level, and accuracy, providing valuable insights into participants' cognitive abilities. To ensure reliable findings, the study implements a rigorous registration process, a lifestyle questionnaire, and tailored difficulty levels.

The research team expects that this study will significantly contribute to our understanding of the influence of sound frequency on memory and attention. Moreover, the developed game holds potential for future advancements and further research. It can serve as a valuable tool for assessing memory span, attention level, and specific processing skills in response to different stimuli. By continually refining the game and conducting longitudinal studies, researchers can gain deeper insights into the long-term effects of sound frequency on cognition.

Furthermore, the data collected from the game can be analysed to identify patterns and correlations, potentially leading to personalized interventions or therapies aimed at enhancing memory and attention. The game's potential for uncovering individual differences in cognitive processes opens avenues for targeted interventions.

7.2.Future Scope of the work

Overall, the future work involving this game holds promising prospects for advancing our knowledge of the relationship between sound frequency and cognitive processes. By expanding the research to larger and more diverse populations, researchers can improve the generalizability of the findings. This could have practical applications in fields such as education, neurorehabilitation, and cognitive enhancement. With continued research and refinement, the game-based psychometric test has the potential to contribute to the development of evidence-based interventions for optimizing memory and attention.

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